

Problem Set 1

(BUSFIN 8200: Asset Pricing Foundations)

Problem Set Rules:

- Your problem set solution must be delivered in the form of a typed report in pdf (not handwritten).
- You must also provide a Git repository with all your files and results. It includes the source files used to generate the pdf (e.g., the .tex script) as well as all coding scripts used to produce the results (e.g., the .R or .py scripts). I need to be able to reproduce your results and pdf from the source files in your Git repository.
- You are allowed to consult any material while working on the problem set (except solutions created by individuals not currently taking the course).
- You are allowed to talk with anyone while working on your problem set. In particular, I highly recommend that you talk with your classmates about the problem set. Research requires collaboration. It is fine if you exchange information and even jointly produce parts of the problem set codes/derivations. However, you have to submit your own solutions.
- Do not abuse the rule above. While you can work together to produce the problem set solution (or even use the solution of a classmate as an initial point in developing your own solution), you cannot simply copy a question solution from a classmate.
- You can also use AI when working on your problem set. However, you need to follow the rules in the “BUSFIN 8200 Problem Sets: AI Policy” document.

Datasets

(EQ Dataset for Questions 1 and 2) The file “EQ Dataset.csv” has the following columns:

- YEAR: the year associated with the observation
- MONTH: the month associated with the observation
- dp: the log of the dividend-to-price ratio ($dp_t = \log(D_t/P_t)$), with D_t reflecting the sum of the dividends paid by the total US equity market (i.e., firms available on CRSP) over the 12 months ending in month t . Dividends account for M&A paid in cash following the procedure described in Gonçalves (2021a)
- dg: the log of the annual dividend growth ($\Delta d_t = \log(D_t/D_{t-12})$), with D_t reflecting the sum of the dividends paid by the total US equity market (i.e., firms available on CRSP) over the 12 months ending in month t . Dividends account for M&A paid in cash following the procedure described in Gonçalves (2021a)
- rf: the log of the annual (deflated) risk-free return ($r_{f,t} = \log(R_{f,t})$) ending in the given month
- re: the log of the annual (deflated) equity market return ($r_{e,t} = \log(R_{e,t})$) ending in the given month

Note: the dataset contains monthly observations of annual variables. So, the empirical analyses in Questions 1 and 2 have to be done accordingly.

(CZ Dataset for Question 3) Download the firm-level signal dataset from Chen and Zimmermann (2022) (available under <https://www.openassetpricing.com/>), which is referred to as “CZ Dataset” in this problem set. For Question 3, you only need the columns “permno” (which is the CRSP stock identifier), “yyyymm” (which identifies the month), “Mom12m” (which is the stock return from month $\tau - 12$ to month $\tau - 1$ from Jegadeesh and Titman (1993)), “BMdec” (which is the log of the book-to-market ratio from Fama and French (1992)), and “GP” (which is the gross profitability variable from Novy-Marx (2013)).

(Dur Dataset for Questions 3) Download the firm-level equity duration dataset from Gonçalves (2021b) (available under <https://andreigoncalves.com/published-papers/>), which is referred to as “Dur Dataset” in this problem set. Merge the CZ Dataset with the Dur Dataset, which contains firm-year equity duration information (column “PERMNO” is the stock identifier, column “FF.YEAR” is the year identifier, and column “Dur” contains the respective equity duration). The timing of the *Dur* variable matches the timing

of the BM variable (so, it is calculated at the end of June of year t using market equity from December of $t - 1$ and accounting information from the end of the fiscal year $t - 1$).

(Bond Dataset for Question 4) Download CRSP Fama-Bliss bond yields from June/1952 to the most recent date for bond strips with maturities from 1 to 5 years.¹ We refer to this file as the “Bond Dataset” in this problem set.

¹In WRDS, go to [CRSP/Annual Update/Treasuries/Fama Bliss Discount Bonds \(Monthly Only\)](#). Then, download the entire file and obtain bond yields from column “TMYTM”. The “TTERMLBL” column provides the bond description and the “MCALDT” column provides the date. We focus on “TTERMLBL = Fama Bliss Discount Bonds - X-Year (Nominal)” for X from 1 to 5.

1 Question 1

1a) Consider the Campbell and Shiller (1989) log-linear (approximate) present value identity for equities

(h is in years):

$$dp_t \approx \frac{\kappa_0 \cdot (\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} + \kappa^H \cdot dp_{t+H} \quad (1.1)$$

$$= \frac{\kappa_0 \cdot (\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \cdot \mathbb{E}_t[dp_{t+H}] \quad (1.2)$$

$$= \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \quad (1.3)$$

where $\kappa = 1/(1 + e^{\bar{dp}})$ and $\kappa_0 = -\log(\kappa) - (1 - \kappa) \cdot \log(1/\kappa - 1)$.

Starting from $R_{e,t+1} = (P_{t+1} + D_{t+1})/P_t$, derive the (approximate) Equations 1.1, 1.2, and 1.3. Explain and justify any assumptions you make.

1b) From Equation 1.1, we have

$$1 = \underbrace{\frac{\text{Cov}\left[dp_t, \sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h}\right]}{\text{Var}[dp]}}_{b_{re}^{(H)}} + \underbrace{\frac{\text{Cov}\left[dp_t, -\sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h}\right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(H)}} + \underbrace{\frac{\text{Cov}[dp_t, \kappa^H \cdot dp_{t+H}]}{\text{Var}[dp]}}_{b_{dp}^{(H)}} \quad (1.4)$$

Applying Equation 1.4 to the EQ Dataset, reproduce the figure provided on slide 1.8 of the Module 1 lecture notes. This figure has H in the x-axis and $(b_{re}^{(H)}, b_{\Delta d}^{(H)}, b_{dp}^{(H)})$ in the y-axis. Explain what the inference from this figure is.

1c) Now, use Equation 1.4 and the EQ Dataset to reproduce a figure analogous to the one in Question (1b) but with the terms from Equation 1.4 implied by the VAR (estimated by OLS)

$$z_{t+1} = \Gamma_0 + \Gamma z_t + \tilde{z}_{t+1} \quad \text{where} \quad \tilde{z}_{t+1} \sim IID(0, \Sigma) \quad (1.5)$$

where $z_t = [\Delta d_t \ r_{e,t} \ dp_t]$

1d) Taking $H \rightarrow \infty$, Equation 1.4 becomes

$$1 = \underbrace{\frac{\text{Cov}\left[dp_t, \sum_{h=1}^{\infty} \kappa^{h-1} \cdot r_{e,t+h}\right]}{\text{Var}[dp]}}_{b_{re}^{(\infty)}} + \underbrace{\frac{\text{Cov}\left[dp_t, -\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \Delta d_{t+h}\right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(\infty)}} \quad (1.6)$$

Using the VAR in Equation 1.5, calculate the values for $b_{re}^{(\infty)}$ and $b_{\Delta d}^{(\infty)}$. Interpret the results.

1e) Gao and Martin (2021) argue that while the Campbell and Shiller (1989) approximation is accurate on average, it can be inaccurate when dp_t is too far from its mean (which happened at some important historical periods such as the late 1990s). They propose the following alternative log-linear (approximate) present value identity:

$$dy_t = (1 - \kappa) \cdot \left(\sum_{h=1}^H \kappa^{h-1} \cdot r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \cdot \Delta d_{t+h} \right) + \kappa^H \cdot dy_{t+H} \quad (1.7)$$

$$= (1 - \kappa) \cdot \left(\sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \right) + \kappa^H \cdot \mathbb{E}_t[dy_{t+H}] \quad (1.8)$$

$$= (1 - \kappa) \cdot \left(\sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \cdot \mathbb{E}_t[\Delta d_{t+h}] \right) \quad (1.9)$$

where $dy_t = \log(1 + D_t/P_t)$ and $\kappa = e^{-\bar{dy}}$.

Starting from $R_{e,t+1} = (P_{t+1} + D_{t+1})/P_t$, derive the (approximate) Equations 1.7, 1.8, and 1.9. Explain and justify any assumptions you make.²

²The steps to derive Equation 1.7 are analogous to the steps you used to derive Equation 1.1 in Questions (1a):

1. Show that $r_{e,t+1} - \Delta d_{t+1} = dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}})$
2. Show that a first-order Taylor approximation yields $\log(1 - e^{-dy_t}) \approx \log(1 - \kappa) + (\kappa/(1 - \kappa)) \cdot (dy_t - \bar{dy})$
3. Substitute the expression from step 2 into the expression from step 1 to obtain $dy_t = (1 - \kappa) \cdot (r_{e,t+1} - \Delta d_{t+1}) + \kappa \cdot dy_{t+1}$
4. Recursively substitute dy_{t+h}

2 Question 2

2a) Using the EQ Dataset, estimate the regression (by OLS)

$$\frac{1}{H} \cdot \sum_{h=1}^H xR_{e,t+h} = a^{(H)} + b^{(H)} \cdot D_t/P_t + \varepsilon_t^{(H)} \quad (2.1)$$

for $H = 1, 2, 3, \dots, 14, 15$ years (with $xR_{e,t} = e^{r_{e,t}} - e^{r_{f,t}}$ and $D_t/P_t = e^{d_{pt}}$). Plot a graph with H in the x-axis and the R_{adj}^2 of these regressions in the y-axis. Describe the results you observe.

2b) Estimate the regression (by OLS)

$$xR_{e,t+1} = a + b \cdot D_t/P_t + \varepsilon_t \quad (2.2)$$

but this time also obtain standard errors for $\hat{b}$ using five different methods:

- (i) The baseline OLS standard errors
- (ii) The White (1980) standard errors
- (iii) The Newey and West (1987) standard errors with 11 lags
- (iv) The Hansen and Hodrick (1980) standard errors with 11 lags
- (v) The Newey and West (1987, 1994) standard errors

The first two methods do not account for autocorrelation. The next two account for autocorrelation up to 11 lags (since $t + 1$ is one year ahead of t and we have a monthly dataset, our overlapping regressions induce artificial autocorrelation up to 11 lags). The last one accounts for autocorrelation with a data-driven method to select the number of lags. Report $\hat{b}$ as well as its t-statistic based on each of the five standard error estimation methods above.³

³We discussed standard errors for time-series regressions in class, but did not cover the details. However, the implementation of these estimators is simple (and available in standard econometric packages). Let's change the time index in our dataset to $\tau = 1, 2, \dots, T$, representing months (as opposed to $t = 1, 2, \dots, T$, which represents years). Then, letting $y = xR_e$, $x_\tau = [1, D/P]$, and $\theta = [a, b]$, Equation 2.2 can be written as $y_{\tau \rightarrow \tau+H} = \theta' x_\tau + \varepsilon_\tau$ (with $H = 12$ in our case). We estimate θ by OLS and obtain standard errors from

$$\widehat{aVar}[\hat{\theta}] = \frac{1}{T} \cdot \left(\frac{1}{T} \cdot \sum_{\tau=1}^T x_\tau x_\tau' \right)^{-1} \hat{S} \left(\frac{1}{T} \cdot \sum_{\tau=1}^T x_\tau x_\tau' \right)^{-1}$$

with the square root of the diagonal elements of the 2×2 matrix $\widehat{aVar}[\hat{\theta}]$ providing the asymptotic standard errors for $\hat{a}$ and $\hat{b}$. S is called the spectral density matrix and is generally given by $S = \sum_{\ell=-\infty}^{\ell=\infty} \mathbb{E}[\varepsilon_\tau x_\tau x_{\tau-\ell}' \varepsilon_{\tau-\ell}]$ (we will learn more about S and its estimators in Module 2 when covering the Generalized Method of Moments). In the absence of autocorrelation and heteroskedasticity, we can use $\hat{S}_{OLS} = \hat{\sigma}_\varepsilon^2 \cdot (1/T \cdot \sum_{\tau=1}^T x_\tau x_\tau')$, which is the baseline OLS estimator. Under heteroskedasticity but no autocorrelation, we can use $\hat{S}_{White} = 1/T \cdot \sum_{\tau=1}^T \hat{\varepsilon}_\tau x_\tau x_\tau' \hat{\varepsilon}_\tau$, which is the White (1980) estimator. If there is autocorrelation and heteroskedasticity, then we typically use a HAC (heteroskedasticity and autocorrelation consistent) estimator, which has the form $\hat{S}_{HAC} = \hat{\Omega}_0 + \sum_{\ell=1}^L w_\ell \cdot [\hat{\Omega}_\ell + \hat{\Omega}_\ell']$, where $\hat{\Omega}_\ell = 1/(\mathcal{T} - \ell) \cdot \sum_{\tau=\ell+1}^T \hat{\varepsilon}_\tau x_\tau x_{\tau-\ell}' \hat{\varepsilon}_{\tau-\ell}$ and L is the number of autocorrelation lags accounted for (see Andrews (1991)). Newey and West (1987) proposes the use of the weighting function $w_\ell = 1 - \ell/(L+1)$, which is the most commonly used HAC estimator and works well for autocorrelation that arises naturally in the data (Newey

- 2c)** Reestimate the regression in Equation 2.2 but this time using the method in Amihud and Hurvich (2004).⁴ Report the b estimate obtained in this question and contrast it to the b estimated obtained in Question (2b). Explain why it is natural for the two estimates to differ.
- 2d)** Estimate the regression in Equation 2.2 again, but this time using out-of-sample (OS) estimation with an expanding window (i.e., using time-varying a_t and b_t that are estimated with historical data). Initiate your OS estimation in December/1940 (i.e., D_t/P_t in December/1939 and $xR_{e,t+1}$ in December/1940). Label the historical (expanding window) mean as $\overline{xR}_{e,t}$, the fitted values from Question (1b) as $\hat{\mathbb{E}}_t^{IS}[xR_e]$, and the OS forecasts from this question as $\hat{\mathbb{E}}_t^{OS}[xR_e]$. Plot the time-series of $\overline{xR}_{e,t}$, $\hat{\mathbb{E}}_t^{IS}[xR_e]$, and $\hat{\mathbb{E}}_t^{OS}[xR_e]$ from December/1940 to the end of the sample. Then, calculate and report the R_{OS}^2 value using the forecasting errors from December/1940 to the end of the sample (note that we calculate R_{OS}^2 instead of $\overline{R}_{OS}^2$ because a degree of freedom adjustment is not needed in an OS analysis). Now, to understand how R_{OS}^2 varies over time, calculate R_{OS}^2 using a 50-year rolling window for the forecasting errors (without modifying the a_t and b_t calculations) and provide a time-series plot of these R_{OS}^2 values (note that the time-series will start in December/1990 and end at the end of your sample). We are using a 50-year rolling window instead of a shorter window because we need a lot of data to calculate meaningful R_{OS}^2 values (otherwise they pick up a lot of noise).
- 2e)** Repeat Question (2d). However, this time estimate the OS coefficients using $\hat{a}_t = \overline{G}_t - 1$ and $\hat{b}_t = \overline{G}_t$ when obtaining $\hat{\mathbb{E}}_t^{OS}[xR_e]$, where $\overline{G}_t$ is the historical (expanding window) average of $e^{\Delta d_t}$ over the same sample used to calculate $\overline{xR}_{e,t}$. These restrictions on a_t and b_t are motivated by a steady state valuation model (with 0% real interest rate), which implies $\mathbb{E}[xR_e] = (G - 1) + G \cdot D/P$.⁵

and West (1994) recommends using $L = \text{const} \cdot T^{1/3}$, with the optimal *const* estimator provided in their equation 2.2). Hansen and Hodrick (1980) propose the use of $w_\ell = 1$ with $L = \text{Overlap} = H - 1$, which better captures the nature of the autocorrelation induced by overlapping regressions (but has the issue that it sometimes leads to a $\widehat{\text{aVar}}[\hat{\theta}]$ that is not positive definite). Hodrick (1992) points out that if $y_{\tau \rightarrow \tau+H}$ reflects the sum of y_τ from $\tau + 1$ to $\tau + H$, then inference using $w_\ell = 1$ with $L = H - 1$ is asymptotically equivalent to inference based on a regression of $y_{\tau+1}$ onto $\sum_{\ell=\tau-(H-1)}^{\ell=\tau} x_\ell$, which leads to better finite sample properties. However, we cannot use the Hodrick (1992) approach since our $y_{\tau \rightarrow \tau+H}$ is an annual return, and thus not the sum of monthly returns (in contrast, if $y_{\tau \rightarrow \tau+H}$ was an annual log return, then it would be the sum of monthly log returns).

⁴We discussed Amihud and Hurvich (2004) in class but did not cover the details. You can read Section II of Amihud and Hurvich (2004) for the underlying econometrics if you want, but the implementation of their estimator is simple. First, regress D_{t+1}/P_{t+1} on D_t/P_t to obtain (by OLS) the estimated intercept, $\hat{\theta}$, and the estimated slope, $\hat{\phi}$. The slope estimate is biased in small samples. You can obtain a bias-corrected slope estimate as $\hat{\phi}^c = \hat{\phi} + (1/T) \cdot (1 + 3 \cdot \hat{\phi}) + (1/T^2) \cdot 3 \cdot (1 + 3 \cdot \hat{\phi})$, where T is the total number of years in the dataset, not the total number of months since $t + 1$ is one year ahead of t . Then, you calculated the bias-corrected residual estimates as $\hat{u}_{t+1}^c = D_{t+1}/P_{t+1} - (\hat{\theta} + \hat{\phi}^c \cdot D_t/P_t)$. Finally, you estimate (by OLS) the regression

$$xR_{e,t+1} = a + b \cdot D_t/P_t + b_u \cdot \hat{u}_{t+1}^c + \varepsilon_t \quad (2.3)$$

We have that $\hat{b}_u$ converges to zero in probability so that $\hat{b}$ converges to b in probability. More importantly, $\hat{b}$ represents an unbiased estimator for b (even in relatively small samples). Amihud and Hurvich (2004) also provide a way to compute standard errors for $\hat{b}$ and to generalize their estimator to multiple predictors. However, in this question you do not need to compute the standard errors and only has one predictor.

⁵This steady state valuation model is not as unrealistic as it seems. For instance, if dividend growth is unpredictable and

3 Question 3

In this question, we work with the time index $\tau = 1, 2, \dots, \mathcal{T}$, representing months (as opposed to $t = 1, 2, \dots, T$, which represents years).

- 3a)** Construct the momentum signal (*MOM*) monthly for the firms with data available in CRSP.⁶ Focus on the period from June of 1963 to the recent date to match the book-to-market construction of Questions (3b). As it is standard in the literature, restrict the analysis to common stocks of firms incorporated in the United States (CRSP shrcd = 10 or 11) trading on NYSE, Amex, or Nasdaq (CRSP exchcd = 1, 2 or 3), and exclude utilities ($4900 \leq \text{SIC} \leq 4949$) and financials ($6000 \leq \text{SIC} \leq 6999$).

To validate your *MOM* construction, merge your dataset with the CZ Dataset and define $MOM_{CZ} = \text{Mom12m}$ (use the column “permno” to identify the stock and the column “yyyymm” to identify the date). For each month in the sample, estimate a cross-firm regression of MOM_{CZ} onto your *MOM* and record the intercept and slope coefficients as well as the R^2 from these regressions. Provide three time series plots, one for the intercepts, one for the slopes, and one for the R^2 values.

- 3b)** Construct the book-to-market signal (*BM*) monthly for the firms with data available in the CRSP and COMPUSTAT datasets.⁷ To ensure all information used in the construction of *BM* is available to investors (see Fama and French (1992)), construct *BM* as of the end of June of year t using CRSP market equity data from December of calendar year $t - 1$ (for *ME*) and COMPUSTAT accounting data from the fiscal year ending in calendar year $t - 1$ (for *BE*).⁸ Then, keep *BM* fixed from June of year t to May of year $t + 1$, updating *BM* again at June of year $t + 1$. Also exclude firm-month observations with $BE \leq 0$. To avoid backfilling bias (see Fama and French (1993)), require a minimum of two previous years in COMPUSTAT for a company to be included in the analysis and construct *BM* from June of 1963 to the most recent date so that the first fiscal year of COMPUSTAT data is 1962. As it is standard in the literature, restrict the analysis to common stocks of firms incorporated in the United States (CRSP shrcd = 10 or 11) trading on NYSE, Amex, or Nasdaq (CRSP exchcd =

expected returns are 100% persistent (i.e., $\mathbb{E}_t[\Delta d_{t+h}] = g$ and $\mathbb{E}_t[r_{e,t+h}] = \mathbb{E}_t[r_e]$), then Equation 1.9 implies $\log(1 + D_t/P_t) = \mathbb{E}_t[r_e] - g$, which is equivalent to $e^{\mathbb{E}_t[r_e]} = e^g \cdot (1 + D_t/P_t)$. Then, subtracting $e^{rf} = 1$ on both sides of the equation leads to $e^{\mathbb{E}_t[r_e]} - e^{rf} = (e^g - 1) + e^g \cdot D_t/P_t$. This equation implies the same type of restrictions on a_t and b_t as the steady-state model does, but in the context of a model in which expected returns (and thus D_t/P_t) are allowed to vary over time.

⁶*MOM* for a stock at the end of month τ refers to the cumulative stock return from month $\tau - 12$ to month $\tau - 1$.

⁷You will need to merge CRSP and COMPUSTAT. The easiest way to do this is to directly use the “CRSP/COMPUSTAT Merged” dataset available on WRDS under CRSP.

⁸We have $ME = |PRC| \cdot SHROUT$ (from CRSP). We have $BE = SE + TXDITC - BVPS$ (from COMPUSTAT). *SE* reflects the stockholders' equity and is given by *SEQ*, *CEQ* + *PSTK*, or *AT* - *LT*, in this sequence of availability. *TXDITC* reflects deferred taxes and is given by the *TXDITC* if it is available (it is set to 0 otherwise). *BVPS* reflects the book value of preferred shareholders and is given by *PSTKRV*, *PSTKL*, or *PSTK*, in this sequence of availability.

1,2 or 3), and exclude utilities ($4900 \leq \text{SIC} \leq 4949$) and financials ($6000 \leq \text{SIC} \leq 6999$).

To validate your BE/ME construction, merge your dataset with the CZ Dataset and calculate $BM_{CZ} = \exp(\text{BMdec})$ (use the column “permno” to identify the stock and the column “yyyymm” to identify the date). For each month in the sample, estimate a cross-firm regression of BM_{CZ} onto your BM and record the intercept and slope coefficients as well as the R^2 from these regressions. Provide three time series plots, one for the intercepts, one for the slopes, and one for the R^2 values.

3c) Let $GP_{CZ} = GP$. Then, for each of the three signals (BM_{CZ} , MOM_{CZ} , and GP_{CZ}), construct decile portfolios as described in the lectures (returns come from CRSP).⁹ Create five types of decile portfolios (for each signal):

- (i) Value-weighted, rebalanced annually (at June), and with NYSE breakpoints
- (ii) Equal-weighted, rebalanced annually (at June), and with NYSE breakpoints
- (iii) Value-weighted, rebalanced monthly, and with NYSE breakpoints
- (iv) Value-weighted, rebalanced annually (at June), and with general breakpoints
- (v) Equal-weighted, rebalanced monthly, and with general breakpoints

For each of the five types of decile portfolios, provide a scatterplot with the decile value (1 to 10) in the x-axis and the average portfolio excess return in the y-axis.¹⁰ To keep the three signals in the same scatterplot (so that you have a total of five scatterplots), use the color blue for BM_{CZ} portfolios, red for MOM_{CZ} portfolios, and green for GP_{CZ} portfolios. Moreover calculate the HML portfolio (decile 10 - decile 1) in each case (you should have a total of 15 HML portfolios) and report in a table the HML average returns as well as their respective t-statistics (using Newey and West (1987, 1994) standard errors, see Footnote 3).

3d) Estimate Fama-MacBeth regressions of $xR_{j,\tau+1}$ on signals available at month τ and report the results in a table (including the t-statistic of each coefficient). Estimate the following regression specifications (at each τ , use both OLS and WLS with month τ market-equity weights):

$$(i) \quad xR_{j,\tau+1} = a + b_{BM} \cdot Q_{j,\tau}^{BM} + \varepsilon_{j,\tau+1}$$

⁹Note that you are using only the signals constructed by Chen and Zimmermann (2022) in this Question (3c). This approach helps you to debug your code since any unusual pattern you observe must come from potential errors in the formation of the portfolios, not from the potential errors in the construction of the signals.

¹⁰The excess return (xR) in Questions (3c) to (3e) refer to the return of a stock or portfolio in excess of the return of a risk-free benchmark. You can obtain returns for a risk-free benchmark in the Ken French data library (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html) under the column “RF” from the file that can be downloaded in the link under “Fama/French 3 Factors”.

- (ii) $xR_{j,\tau+1} = a + b_{GP} \cdot Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}$
- (iii) $xR_{j,\tau+1} = a + b_{Dur} \cdot Q_{j,\tau}^{Dur} + \varepsilon_{j,\tau+1}$
- (iv) $xR_{j,\tau+1} = a + b_{BM} \cdot Q_{j,\tau}^{BM} + b_{GP} \cdot Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}$
- (v) $xR_{j,\tau+1} = a + b_{Dur} \cdot Q_{j,\tau}^{Dur} + b_{BM} \cdot Q_{j,\tau}^{BM} + \varepsilon_{j,\tau+1}$
- (vi) $xR_{j,\tau+1} = a + b_{Dur} \cdot Q_{j,\tau}^{Dur} + b_{GP} \cdot Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}$
- (vii) $xR_{j,\tau+1} = a + b_{Dur} \cdot Q_{j,\tau}^{Dur} + b_{BM} \cdot Q_{j,\tau}^{BM} + b_{GP} \cdot Q_{j,\tau}^{GP} + \varepsilon_{j,\tau+1}$

where $Q_{j,\tau}^X$ reflects the quantile of signal $X_{j,\tau}$ in the cross-firm distribution of X at month τ (for BM and GP you should use the Chen and Zimmermann (2022) version of the signals). The use of quantiles deals with the fact that signals often have skewed distributions and outliers.¹¹

3e) Gonçalves (2021b) proposes the use of multivariate regressions for portfolios analogous to the use of firm-level multivariate regressions in Question (3d). For instance, if we have 10 BM_{CZ} decile portfolios and 10 GP_{CZ} decile portfolios, then for each portfolio p we can calculate the average BM_{CZ} decile of its firms (let's call it $Dec_{p,\tau}^{BM}$) as well as the average GP_{CZ} decile of its firms (let's call it $Dec_{p,\tau}^{GP}$) and estimate the panel regression $xR_{p,\tau+1} = a + b_{BM} \cdot Dec_{p,\tau}^{BM} + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$ by pooled OLS using the 20 portfolios with Driscoll and Kraay (1998) standard errors.¹² Following this approach, estimate the following regression specifications and report the results in a table (including t-statistics):

- (i) $xR_{p,\tau+1} = a + b_{BM} \cdot Dec_{p,\tau}^{BM} + \varepsilon_{p,\tau+1}$
- (ii) $xR_{p,\tau+1} = a + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$
- (iii) $xR_{p,\tau+1} = a + b_{Dur} \cdot Dec_{p,\tau}^{Dur} + \varepsilon_{p,\tau+1}$
- (iv) $xR_{p,\tau+1} = a + b_{BM} \cdot Dec_{p,\tau}^{BM} + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$
- (v) $xR_{p,\tau+1} = a + b_{Dur} \cdot Dec_{p,\tau}^{Dur} + b_{BM} \cdot Dec_{p,\tau}^{BM} + \varepsilon_{p,\tau+1}$
- (vi) $xR_{p,\tau+1} = a + b_{Dur} \cdot Dec_{p,\tau}^{Dur} + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$
- (vii) $xR_{p,\tau+1} = a + b_{Dur} \cdot Dec_{p,\tau}^{Dur} + b_{BM} \cdot Dec_{p,\tau}^{BM} + b_{GP} \cdot Dec_{p,\tau}^{GP} + \varepsilon_{p,\tau+1}$

Estimate each specification above using value-weighted portfolios as well as equal-weighted portfolios (in both cases the portfolios are rebalanced annually and rely on NYSE breakpoints). Moreover, for the BM and GP portfolios use the Chen and Zimmermann (2022) version of the signals.

¹¹An alternative approach is to use the cross-firm z-score of winsorized log X , but this log approach cannot be used for signals that can take negative values (not the case here, but often the case in other applications).

¹²Gonçalves (2021b) shows that this panel regression of portfolio returns on portfolio declines is a natural generalization of the HML average returns often emphasized in the literature. For instance, the coefficient estimate from a univariate regression of portfolio returns on portfolio deciles is equivalent to a weighted average of HML average returns, where HML portfolios are obtained from $Decile_{10} - Decile_1$, $Decile_9 - Decile_2$, $Decile_8 - Decile_3$, $Decile_7 - Decile_4$, and $Decile_6 - Decile_5$.

4 Question 4

4a) The “Bond Dataset” contains yields in percentual terms, $Y_{b,t}^{(H)} \cdot 100\%$. Construct time series of:

- (i) Log yields: $y_{b,t}^{(H)} = \log(1 + Y_{b,t}^{(H)})$
- (ii) Log forward rates: $f_{b,t}^{(H)} = H \cdot y_{b,t}^{(H)} - (H - 1) \cdot y_{b,t}^{(H-1)}$
- (iii) Log annual returns: $r_{b,t}^{(H)} = H \cdot y_{b,t-1}^{(H)} - (H - 1) \cdot y_{b,t}^{(H-1)}$

For $H = 2, 3, 4, 5$, calculate and report in a table the average values of¹³

- (i) $xy_{b,t}^{(H)} = y_{b,t}^{(H)} - y_{b,t}^{(1)}$
- (ii) $xf_{b,t}^{(H)} = f_{b,t}^{(H)} - f_{b,t}^{(1)}$
- (iii) $xr_{b,t}^{(H)} = r_{b,t}^{(H)} - r_{b,t}^{(1)}$

4b) For $H = 2, 3, 4, 5$, estimate the regression

$$(1/H) \cdot xr_{b,t \rightarrow t+H}^{(H)} = a^{(H)} + b^{(H)} \cdot xy_{b,t}^{(H)} + \varepsilon_t^{(H)} \quad (4.1)$$

where the hold-to-maturity excess return is $xr_{b,t \rightarrow t+H}^{(H)} = \sum_{h=1}^H xr_{t+h}^{(H-h+1)} = \sum_{h=1}^H (r_{t+h}^{(H-h+1)} - r_{t+h}^{(1)})$.¹⁴

Using the results from the regressions in Equation 4.1, reproduce a table analogous to the first panel of the first table on slide 5.5 of the Module 1 lecture notes. For the t-statistics, use the method in Hansen and Hodrick (1980) (see Footnote 3).

4c) For $H = 2, 3, 4, 5$, estimate the regression¹⁵

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} \cdot xf_{b,t}^{(H)} + \varepsilon_t^{(H)} \quad (4.2)$$

Using the results from the regressions in Equation 4.2, reproduce a table analogous to the second panel of the first table on slide 5.5 of the Module 1 lecture notes. For the t-statistics, use the method in Newey and West (1987, 1994) (see Footnote 3).

¹³Note that the excess returns are not deflated, but would be identical if deflated. In particular, the deflated log return of a bond is $\log(R_{b,t}^{(H)} \cdot \Pi_t / \Pi_{t-1}) = r_{b,t}^{(H)} - \Delta\pi_t$. As such, deflated excess log return of a bond is $xr_{b,t}^{(H)} = (r_{b,t}^{(H)} - \Delta\pi_t) - (r_{b,t}^{(1)} - \Delta\pi_t) = r_{b,t}^{(H)} - r_{b,t}^{(1)}$.

¹⁴Note that $xr_{b,t \rightarrow t+H}^{(H)} \neq \sum_{h=1}^H (r_{t+h}^{(H)} - r_{t+h}^{(1)})$. The intuition is that we want to capture the hold-to-maturity return on the H -year bond. In one year, this bond will have maturity of $H - 1$. Then, in the subsequent year it will have maturity of $H - 2$ and so on.

¹⁵Note that for $H = 2$ we have $xr_{b,t \rightarrow t+2}^{(2)} = xr_{b,t+1}^{(2)}$ (to verify this result, just substitute $H = 2$ in the expression for $xr_{b,t \rightarrow t+2}^{(2)}$ in question (4b)). Moreover, with $H = 2$ we have $xf_{b,t}^{(2)} = 2 \cdot xy_{b,t}^{(2)}$ (to verify this result, just substitute $H = 2$ in the expression for $xf_{b,t}^{(H)}$ in question (4a)). So, $b^{(2)}$ in Equation 4.2 is the same as $b^{(2)}$ in Equation 4.1 (to verify this result, just substitute $xr_{b,t+1}^{(2)} = xr_{b,t \rightarrow t+2}^{(2)}$ and $xf_{b,t}^{(2)} = 2 \cdot xy_{b,t}^{(2)}$ in Equation 4.2 and divide both sides of the equation by $H = 2$).

4d) Estimate the Cochrane and Piazzesi (2005) factor from

$$\frac{1}{4} \cdot \sum_{H=2}^5 x_{r,b,t+1}^{(H)} = \theta_0 + \underbrace{\sum_{H=1}^5 \theta_h \cdot f_{b,t}^{(H)}}_{cp_t} + u_t \quad (4.3)$$

and provide a time-series plot of cp_t with NBER recession gray areas in the background (the NBER recession dummies are available under <https://fred.stlouisfed.org/series/USREC>)

4e) For $H = 2, 3, 4, 5$, estimate the regression

$$x_{r,b,t+1}^{(H)} = a^{(H)} + b^{(H)} \cdot cp_t + \varepsilon_t^{(H)} \quad (4.4)$$

Using the results from the regressions in Equation 4.4, reproduce a table analogous to the second table on slide 5.5 of the Module 1 lecture notes. For the t-statistics, use the method in Newey and West (1987, 1994) (see Footnote 3).

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